ANTHONY ETIM

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EDUCATION

Yale University, New Haven, CT August 2021 – May 2027
Ph.D. in Electrical Engineering
Research Assistant, Computer Architecture and Security Lab (CASLAB).
Advisor: Prof. Jakub Szefer

Yale University, New Haven, CT August 2021 – May 2024
M.S., M.Phil. in Electrical Engineering

Villanova University, Villanova, PA August 2017 – May 2021
B.S. in Electrical Engineering,
Minors in Computer Science and Computer Engineering

TECHNICAL SKILLS

Programming: Python, C++, C, MATLAB, Java, SQL, VHDL, Verilog, SystemVerilog, Haskell.
Tools: vLLM, AWS, Vivado, Quartus, GitHub, Keras, JAX, PyTorch, TensorFlow, Hugging Face, Git.
Technological Devices: Raspberry Pi, Arduino.
Expertise: Hardware Security, AI Security, FPGA, Quantum Error Correction, Computer Architecture, Machine Learning, Deep Learning, Hardware Architecture, AI HW/SW Co-design, Neural Network Architecture, Large Language Models, Reinforcement Learning.

RESEARCH EXPERIENCE

Google, *Software Engineering Intern*, ML Frameworks, Mountain View, CA June 2026 – August 2026

- Integrated vLLM into KerasHub to serve CausalLM models for TPU inference.
- Routed KerasHub attention through vLLM's paged-attention kernel, raising TPU v6e serving throughput 1.7–7.2× over native generate() across different models under concurrent load.
- Enabled vLLM serving for model families, validated token-for-token parity with generate(), and authored the keras.io vLLM serving guide.

Yale University, *Graduate Researcher*, New Haven, CT August 2021 – Present

- Fortified Tiny ML algorithms — induced near-100% misclassification via voltage and electromagnetic glitches on four models; restored accuracy with random self-reducibility and majority voting.
- Built and evaluated fault-tolerant ML-based quantum readout-correction and error-correction decoders to improve resilience against physical fault-injection attacks.
- Built self-correcting fault-injection defenses — applied randomized perturbations and majority-voting to detect and correct faults in ML and cryptographic accelerators, optimizing robustness via error-distribution analysis.
- Pioneered attacks and defenses in ML accelerators — first to recover full-color inputs from ML accelerators on multi-tenant cloud FPGAs via ML-enhanced TDC side-channel analysis.
- Secured Q-Learning and Deep Q-Learning systems — evaluated voltage-glitch and bit-flip attacks, revealed defense gaps, and developed a fault-resilient implementation for dynamic control environments.

- Engineered adversarial attacks and defenses — deployed sticker perturbations on traffic-sign CNNs (92 % outdoor success) and implemented a "time-traveling" defense using historical image ensembles for 100% robustness against attacks.

Scale AI, *Gen AI Intern,* San Francisco, CA, Remote June 2025 – December 2025

- Designed domain-specific and adversarial evaluations for frontier LLMs; performed failure-mode analyses and delivered recommendations adopted in model updates.
- Oversaw end-to-end training and fine-tuning of a code-reasoning LLM; developed C++/Python solutions for top-3% Olympiad problems, applied LoRA/PEFT to enhance chain-of-thought, and benchmarked Olympiad-level reasoning to surface failure modes that informed architecture improvements.

Villanova University, *Undergraduate Researcher*, Villanova, PA Fall 2020

- Used Matrix Singular Value Decomposition (SVD) technique to optimize the deep neural network (DNN) on the AVNET Ultra96-V2 FPGA development board.
- Evaluated the performance, accuracy, and energy consumption of the optimized system.

Electrical Engineering Intern, National Grid, Albany, NY, Remote Summer 2020

- Collaborated with a team of 4 engineers to help reorganize and plan the grid network using various tools.
- Modelled data to fit various design requirements and constraints of the power system.
- Assisted in the management and creation of a SharePoint Setting Repository for the handoff of smart control settings to the field device engineers.
- Collaborated with other engineers to build a database of DMX and control house plans for the grid network.

Villanova University, *Undergraduate Researcher*, Villanova, PA March 2018 – May 2019

- Conducted research on the development of the flow network modeling tool Villanova Thermodynamic Analysis of Systems (VTAS), which models the energy flows throughout a data center.
- Upgraded the Graphical User Interface (GUI) for the Villanova Thermodynamic Analysis of Systems (VTAS) data center flow network modeling tool.
- Developed the GUI for the VTAS electrical system layout.

PUBLICATIONS

- **Anthony Etim**, Srilalith Nampally, Aubtin Rasouli, Dustin Mazza, Krishna Chilakapati, Tinghung Chiu, Ferhat Erata, Leyla Nazhandali, Wenjie Xiong and Jakub Szefer. "Fault Injection Attacks and Countermeasures on TinyML Algorithms". IEEE International Symposium on Hardware Oriented Security and Trust **(HOST)**, 2026. **Best Student Paper Award.**
- **Anthony Etim** and Jakub Szefer. "Voltage Fault Injection Against On-Device Language Models". International Workshop on Hardware and Architectural Support for Security and Privacy **(HASP)**, 2026.
- **Anthony Etim** and Jakub Szefer. "Power Side-Channel Attacks on Machine-Learning-based Quantum Error Correction Decoders". International Workshop on Hardware and Architectural Support for Security and Privacy **(HASP)**, 2026.

- **Anthony Etim** and Jakub Szefer. "Fault Attacks on ML-based Quantum Control and Error Correction". IEEE International Conference on Quantum Computing and Engineering **(QCE)**, 2026.
- Tom Slooff, **Anthony Etim**, Jiaqi Yu, Francesco Regazzoni and Jakub Szefer. "Fault Injection Attacks on Q-Learning: Analysis, Vulnerabilities, and Countermeasures". In Submission.
- **Anthony Etim**, Kenan Erol and Jakub Szefer. "ML-Assisted Power Side-Channel Attacks on ML Accelerators in Shared FPGAs". In Submission.
- **Anthony Etim** and Jakub Szefer. "Adversarial Attacks on Traffic Sign Classification in Autonomous Driving". In Submission.
- Ferhat Erata, TingHung Chiu, **Anthony Etim**, Srilalith Nampally, Tejas Raju, Rajashree Ramu, Ruzica Piskac, Timos Antonopoulos, Wenjie Xiong, and Jakub Szefer. "Systematic Use of Random Self-Reducibility in Cryptographic Code against Physical Attacks". IEEE International Conference on Computer-Aided Design **(ICCAD)**, 2024.
- **Anthony Etim**, Shanquan Tian, and Jakub Szefer. "Extending FPGA Information Leaks with Trojan Phantom Circuits". IEEE International Symposium on Secure and Private Execution Environment Design **(SEED)**, 2024.
- Theodoros Trochatos, **Anthony Etim**, and Jakub Szefer. "Covert-channels in FPGA-enabled SmartSSDs". ACM Transactions on Reconfigurable Technology and Systems **(TRETS)**, 2024. Presented in the Journal Track of the 22nd International Conference on Field-Programmable Technology **(FPT)**, 2023.

TEACHING EXPERIENCE

- **Teaching Fellow,** Yale University, New Haven, CT — Spring 2023
  Introduction to Computer Engineering (EENG 201)
- **Teaching Fellow,** Yale University, New Haven, CT — Fall 2022
  Introduction to Electronics (EENG 200)

SERVICE

- **IEEE International Symposium on Hardware Oriented Security and Trust (HOST) 2027,** Technical Program Committee
  July 2026
- **Great Lakes Symposium on VLSI (GLSVLSI) 2026,** Hardware Security Program Committee
  March 2026
- **ACM Transactions on Reconfigurable Technology and Systems (TRETS),** Reviewer
  February 2026
- **IEEE Transactions on Information Forensics & Security (T-IFS),** Reviewer
  September 2025
- **ACM Transactions on Architecture and Code Optimization (TACO),** Reviewer
  August 2025
- **Yale Cloud Computing and FPGA Security Symposium (CCFS) 2022,** Co-organizer
  November 2022
- **Yale Grad Society of Women Engineers,** Undergrad Liaison Co-chair
  September 2022 – May 2024

LEADERSHIP EXPERIENCE

Tau Beta Pi, National Engineering Honor Society, Villanova Chapter, Vice-President
March 2020-May 2021

- Planned and conducted various professional events for the chapter's members such as the initiation information session for new members.
- Helped in the advancement of the technical and professional education of the active members by connecting them with various alumni.

Villanova Engineering Student Council, Co-Chair September 2019- May 2021

- Planned events for the College of Engineering and acted as a bridge between the students and faculty.

National Society of Black Engineers, Senator August 2017-September 2018

- Represented Villanova chapter at the 2017 Fall Regional Conference in Greensboro, NC.

SELECTED HONORS AND AWARDS

- Sigma Xi, The Scientific Research Honor Society, Member March 2026
- Meta PhD Forum October 2025
- AsianHOST PhD Forum December 2024
- Yale New Student Fellowship September 2021
- Dean's Award for Academic Excellence May 2021
- Dean's Award for Meritorious Service May 2021
- Dean's List December 2017 - May 2021
- Tau Beta Pi, the National Engineering Honor Society, Spring 2020
  Selected based on academic ranking 1/8th of the junior class.
- Klingler Unitas Prize, April 2020
  Villanova Student Entrepreneurship Competition
- Klingler Unitas Prize, April 2019
  Villanova Student Entrepreneurship Competition